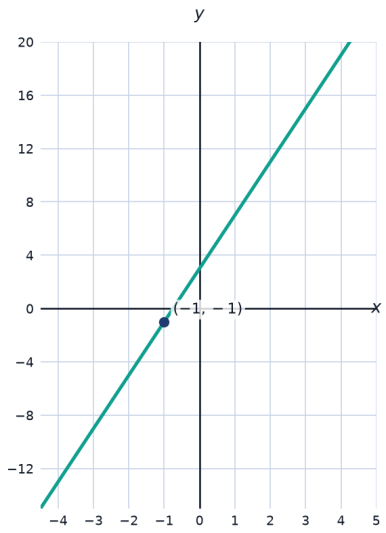
# Exponential & Logarithmic Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

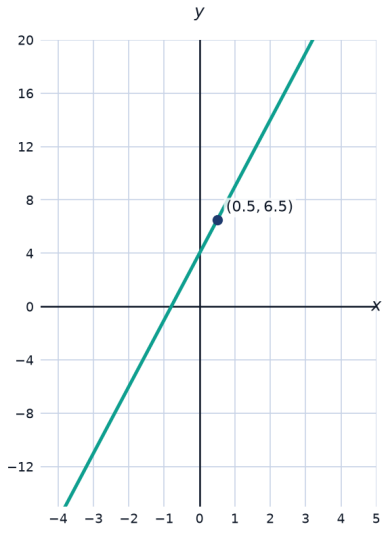
**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(5)$.




**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(2)$.



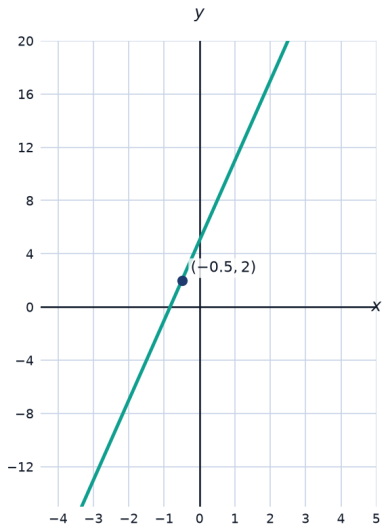

**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

© ClipSAT — clipsat.org | All Rights Reserved

**4** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(2)$.

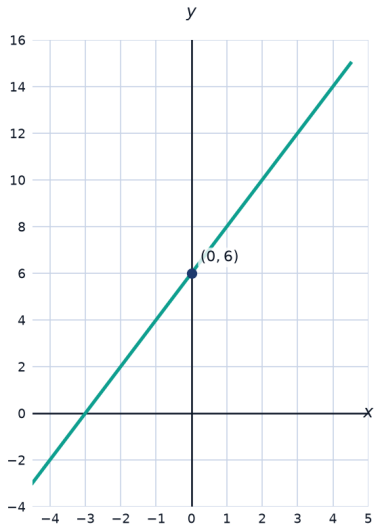



**5** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for patterns with rational and irrational numbers consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

**6** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(2)$.




**7** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for simplifying radical expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

© ClipSAT — clipsat.org | All Rights Reserved

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

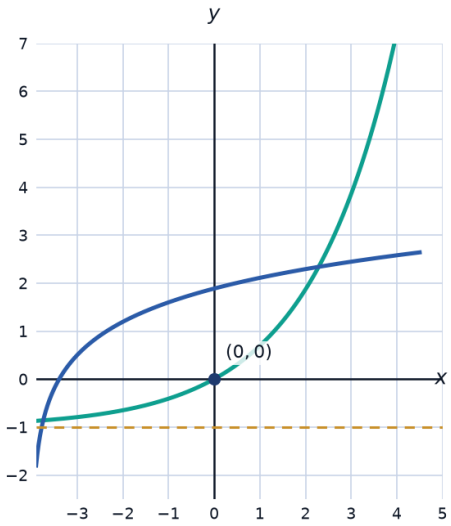
Solve $2^{x+1} = 5^{x-1}$, giving your answer to 3 significant figures.

**9** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the logarithmic equation $\log_3(x^2 - 7) = 2$ for all real solutions, and verify that all roots lie within the valid domain.

**10** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 2 · MediumAn ecologist models the population $P$ of trout in a lake as $P(t) = 1400 - 35t$, where $t$ is the number of years after 2015. What is the best interpretation of the number $35$ in this model?




**11** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedWhich ordered pair $(x, y)$ is a solution to the system of inequalities $y > 2x - 3$ and $y \leq -x + 4$?

**12** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 4 · StrategyA student finishes the first 18 questions of Module 1 with 9 minutes remaining. Questions 19 and 20 are lengthy geometry proofs, while Questions 21 and 22 are quick Desmos system intersection problems. In what order should the student attack the remaining 4 questions, and why?

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumA student solves a linear equation and obtains the fraction $x = -\frac{7}{4}$. What are the valid ways to enter this answer into the Digital SAT Student-Produced Response (SPR) grid?

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsFind the $x$-intercept and $y$-intercept of the linear equation $3x - 6y = 18$.

© ClipSAT — clipsat.org | All Rights Reserved

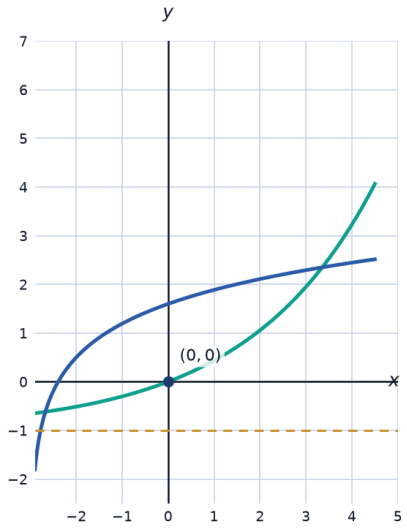
## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Linear Equations & Lines

If $f(x) = 4x - 3$ and $g(x) = x^2 + 1$, find (a) $gf(2)$ and (b) an expression for $g^{-1}(x)$ for $x \geq 1$.

**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Rational Expressions

Level 3 · AdvancedFor the piecewise function $g(x) = \{2x + 5, x \leq -1; x^2 - 3, x > -1$, what is the value of $g(-1) + g(3)$?




**17** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Statistics & Data Analysis

Level 2 · MediumA study of 100 students recorded test results based on exam prep: • Studied: 54 passed, 6 failed (60 total) • Did not study: 18 passed, 22 failed (40 total) What is the probability that a randomly chosen student passed the exam, given that the student studied?

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratic Functions

Level 4 · SynthesisIf $f(x - 3) = 4x + 5$ for all real numbers $x$, what is the value of $f(2)$?

**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Exponents & Polynomials

Level 2 · MediumLet $f(x) = x - 4$ and $g(x) = 2x^2 + 1$. If $h(x) = g(f(x))$, what is the value of $h(1)$?

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Functions & Transformations

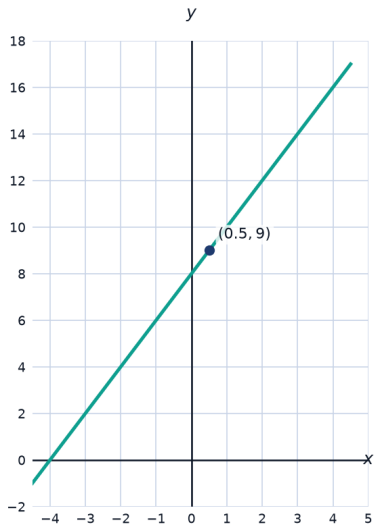
Level 1 · FoundationsFor the function $f(x) = 2x^2 - 3x + 5$, what is the value of $f(-3)$?

© ClipSAT — clipsat.org | All Rights Reserved

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(2)$.




**a.** 12

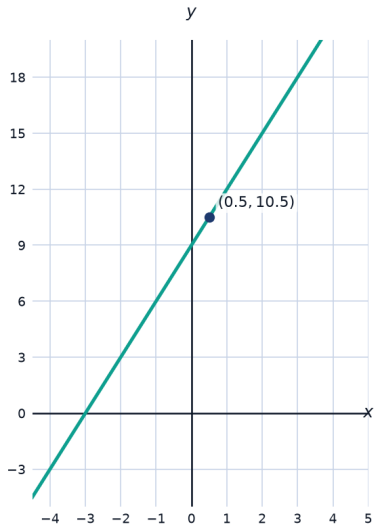
**b.** 14

**c.** 4

**d.** 16

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(2)$.




**a.** 15

**b.** 18

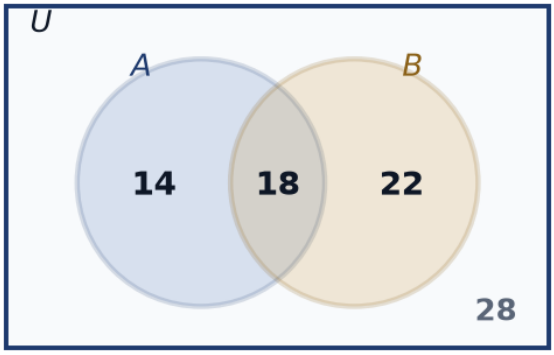
**c.** 6

**d.** 19

© ClipSAT — clipsat.org | All Rights Reserved

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for multiplication properties of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?




**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** 19.5

**b.** 20.5

**c.** 21.5

**d.** 23.5

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function $f$ is defined by $f(x) = a \cdot b^x + c$, where $a$, $b$, and $c$ are constants with $a > 0$, $b > 0$, and $beq1$. In the $xy$-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

**a.** 20

**b.** 24

**c.** 32

**d.** 44

© ClipSAT — clipsat.org | All Rights Reserved

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where $a$ and $k$ are real constants. What is the value of $f(2)$?

**a.** 19

**b.** 31

**c.** 16

**d.** 11

**27** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

**a.** $y = -5$

**b.** $y = 0$

**c.** $y = 3$

**d.** $x = -1$

**28** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of $a$ and the value of $b$.

**a.** a = 4, b = 3

**b.** a = 3, b = 4

**c.** a = 6, b = 2

**d.** a = 2, b = 6

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of $a$.

**a.** 2

**b.** 3

**c.** 4

**d.** 6

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

**a.** 6

**b.** 8

**c.** 9

**d.** 5

© ClipSAT — clipsat.org | All Rights Reserved

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of $y$ when $x = 3$ for the function $f(x) = 2^{x+1}$?

**a.** 4

**b.** 8

**c.** 6

**d.** 16

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the half-life formula for exponential decay?

**a.** $A(t) = A_0(2)^{t/h}$

**b.** $A(t) = A_0e^{-t}$

**c.** $A(t) = A_0/2^t$

**d.** $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$

**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve: $e^{2x} = 5$. Express in terms of $\ln$.

**a.** $x = \ln k**2 + 1 - k$

**b.** $x = \ln \frac{k**2+1}{k}$

**c.** $x = \frac{\ln k**2+1}{k}$

**d.** $x = \frac{k}{\ln k**2+1}$

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

**a.** $A = Pe^{rt}$

**b.** $A = P/e^{rt}$

**c.** $A = P(1 + r)^t$

**d.** $A = Prt$

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The equation $\log_a x = b$ and $\log_b x = a$ simultaneously implies:

**a.** No simultaneous solution

**b.** $a = b$

**c.** $x = 0$

**d.** $x = a^b = b^a$

© ClipSAT — clipsat.org | All Rights Reserved

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve: $5^x = 12$. Give exact answer.

**a.** $x = 12/5$

**b.** $x = \frac{\ln 12}{\ln 5}$

**c.** $x = \ln 12 - \ln 5$

**d.** $x = \log_{12} 5$

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Write using natural log: $\log_b x =$

**a.** $\frac{\ln b}{\ln x}$

**b.** $\ln x - \ln b$

**c.** $\ln x \cdot \ln b$

**d.** $\frac{\ln x}{\ln b}$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratic Functions

The value of $e$ to 4 decimal places is:

**a.** 2.1787

**b.** 3.1416

**c.** 2.6180

**d.** 2.7183

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Exponents & Polynomials

Convert to exponential form: $\log_5 25 = 2$.

**a.** $5^{25} = 2$

**b.** $2^5 = 25$

**c.** $5^2 = 25$

**d.** $25^2 = 5$

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Functions & Transformations

Which is the y-intercept of $f(x) = 5 \cdot 3^x$?

**a.** $(1, 15)$

**b.** $(0, 15)$

**c.** $(0, 5)$

**d.** $(0, 3)$

© ClipSAT — clipsat.org | All Rights Reserved

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find $A/A_0$ at $t = 10$.

**a.** 8

**b.** 2

**c.** 4

**d.** 10

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Sequences & Series

Which of these is equivalent to $e^{2\ln 3}$?

**a.** $2e^3$

**b.** 9

**c.** 6

**d.** $\ln 9$

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Factoring Polynomials

Find the derivative of $f(x) = x^x$.

**a.** $x^x \ln x$

**b.** $x^{x+1}$

**c.** $x \cdot x^{x-1}$

**d.** $x^x(1 + \ln x)$

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Radicals & Exponents

Find the number of solutions to $e^x = x + 2$.

**a.** 1

**b.** 2

**c.** Infinitely many

**d.** 0

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Linear Equations & Lines

Solve: $3^x = 2^{x+1}$. Use logarithms.

**a.** $x = \frac{\ln 2}{\ln 3}$

**b.** $x = 1$

**c.** $x = \frac{\ln 3}{\ln 2}$

**d.** $x = \frac{\ln 2}{\ln 3 - \ln 2}$

© ClipSAT — clipsat.org | All Rights Reserved